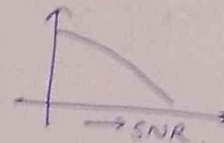


$$BER = \frac{1}{2} \left(1 - \sqrt{\frac{1}{\frac{2}{SNR} - 1}} \right)$$

If $SNR \gg \frac{2}{SNR} \approx 0$

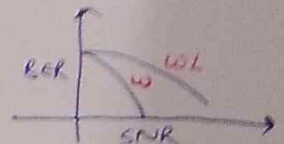
$$BER \approx \frac{1}{2(SNR)}$$



BER (wired)
 $\propto \sqrt{SNR}$

BER (wireless)

$$BER \approx \frac{1}{2 SNR}$$



$$BER = 10^{-4}$$

$$SNR|_{WL} = 10 \times SNR|_w$$

$$SNR = \frac{P}{N_0}$$

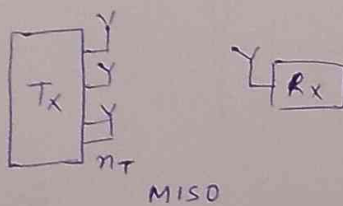
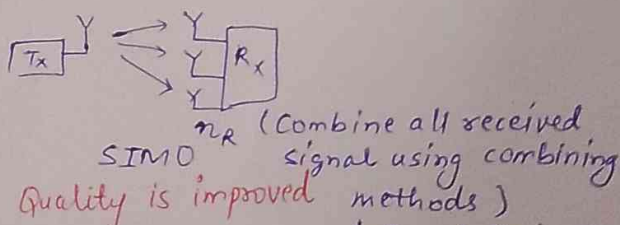
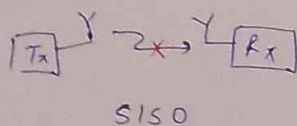
$$SNR = |h|^2 \frac{P}{N_0}$$

fading

How we can overcome this challenge i.e. fading?

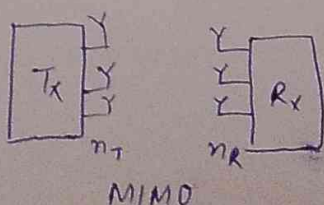
⇓
(*) MIMO

Multiple antenna at T_x and R_x side.



- Equal Gain C. (add with same factor)
- Selection C. (pick the best)
- Maximum Ratio C. (factor such the final signal has max SNR)

- All Antenna transmitting diff data (more data rate)
- If same signal + same data rate → more quality



To achieve both data rate and quality.

* Outage Probability

(Capacity < Req. Rate)

does not
meeting the
requirement

$$Pr(y_i < y_{th})$$

SISO

$$Y \rightarrow Z$$

$$y = hx + n$$

$$n \sim \mathcal{N}(0, \sigma^2)$$

$$x \rightarrow P$$

$h \rightarrow$ fading

$$y_i = |h_i|^2 \frac{P}{\sigma^2}$$

$$P_{out} = Pr\left(|h_i|^2 \frac{P}{\sigma^2} < y_{th}\right)$$

$$= Pr\left(|h_i|^2 < \frac{y_{th}}{\bar{y}}\right); \left[\bar{y} = \frac{P}{\sigma^2}\right]$$

$$a = |h_i| \rightarrow \text{Rayleigh distribution}$$

$$= f_A(a) = 2a e^{-a^2}$$

Note -

$$x \rightarrow f_X(x)$$

$$y = x^2 \rightarrow dy = 2x dx = 2\sqrt{y} dy$$

$$f_Y(y) = ?$$

$$f_Y(y) = \frac{f_X(x) dx}{dy}$$

So,

$$y = |h_i|^2$$

$$f_X(x) = 2x e^{-x^2}$$

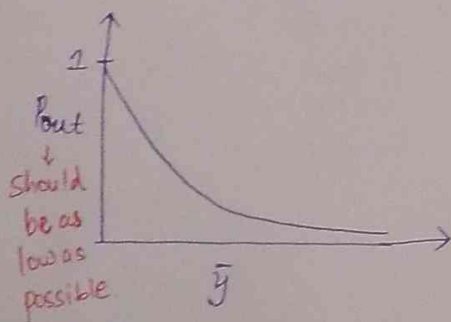
$$f_X(y) = 2\sqrt{y} e^{-y}$$

$$f_Y(y) = \frac{2\sqrt{y} e^{-y} dx}{2\sqrt{y} dy} = e^{-y}$$

$$P_{out} = \int_0^{y_{th}/\bar{y}} f_Y(y) dy$$

$$= \int_0^{y_{th}/\bar{y}} e^{-y} dy$$

$$P_{out} = \left[1 - e^{-\frac{y_{th}}{\bar{y}}}\right]$$



* Channel Capacity

$$C = B \log_2(1 + y_i)$$

$$= B \log_2(1 + |h_i|^2 \bar{y})$$

$y_i \rightarrow$ instantaneous SNR

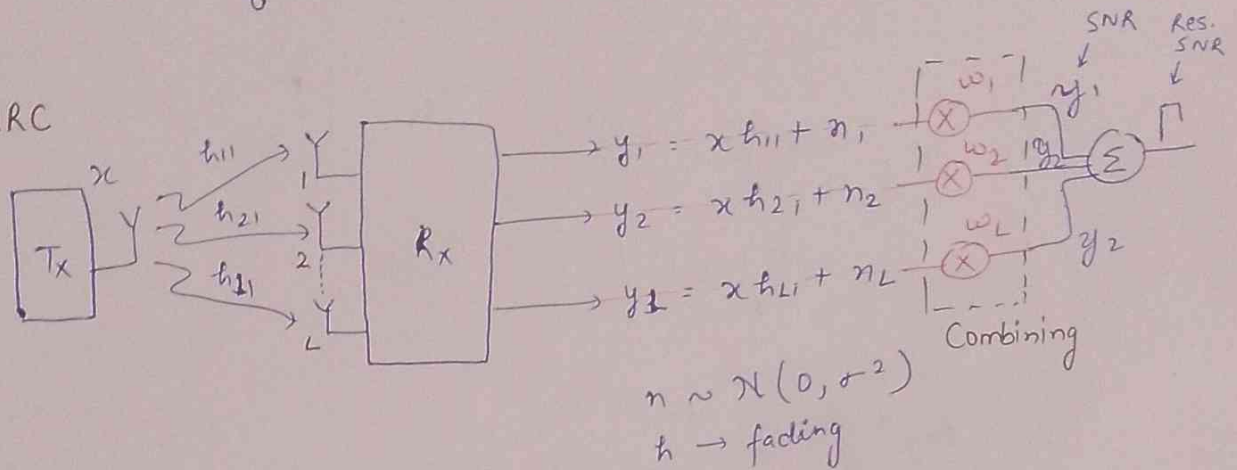
Average capacity, $E[C_{th}] \rightarrow$ ergodic capacity

$$|h|^2 = y$$

$$E[C/R] = \int C f_Y(y) dy$$

$$= \int_0^\infty B \log_2(1 + y \bar{y}) f_Y(y) dy$$

(*) MRC



$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_L \end{bmatrix}_{L \times 1} = x \begin{bmatrix} h_{11} \\ h_{21} \\ \vdots \\ h_{L1} \end{bmatrix}_{L \times 1} + \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_L \end{bmatrix}_{L \times 1}$$

(Choose $w_1, w_2, \dots, w_L$ to get max. Γ)

$$\bar{y} = x \bar{h} + \bar{n}$$

$$\begin{aligned} \hat{y}_1 &= w_1 h_{11} x + w_1 n_1 \\ \hat{y}_2 &= w_2 h_{21} x + w_2 n_2 \\ &\vdots \\ \hat{y}_L &= w_L h_{L1} x + w_L n_L \end{aligned}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_L \end{bmatrix} = x \begin{bmatrix} w_1 & w_2 & \dots & w_L \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{21} \\ \vdots \\ h_{L1} \end{bmatrix} + \begin{bmatrix} w_1 n_1 \\ w_2 n_2 \\ \vdots \\ w_L n_L \end{bmatrix}$$

Rec. signal after combining Schemes applied

$$\Gamma = \frac{|w^H h|^2 P}{|w^H w| \sigma^2}$$

$$; |w^H w| = 1 \quad ; \quad \frac{P}{\sigma^2} \text{ constant}$$

To maximise Γ , max $(|w^H h|^2)$

$$w = k h$$

$$|w|^2 = 1 \Rightarrow |k h|^2 = 1 \Rightarrow k = \frac{1}{|h|}$$

$$|a^H b| = |a||b| \cos \theta$$

$b = k a$ when $\theta = 0$

$$\omega_{\text{opt.}} = \frac{h}{|h|}$$

So,

$$\text{Max. SNR, } \Gamma = \frac{|h|^2 P}{\sigma^2}$$

$$\begin{aligned} \Gamma &= \left[|h_{11}|^2 + |h_{21}|^2 + \dots + |h_{L1}|^2 \right] \frac{P}{\sigma^2} \\ &= \underbrace{\sum_{i=1}^L |h_{i1}|^2}_g \frac{P}{\sigma^2} = g \frac{P}{\sigma^2} \end{aligned}$$

$|h| \rightarrow$ Rayleigh distribution

$|h|^2 \rightarrow$ exponential distribution

$\sum |h|^2 \rightarrow$ chi-square distribution of order L (not exp.)

$\rightarrow$ Consider BPSK Mod.

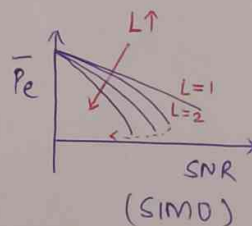
$$P_e = Q(\sqrt{2 \text{SNR}})$$

(Inst. BER)

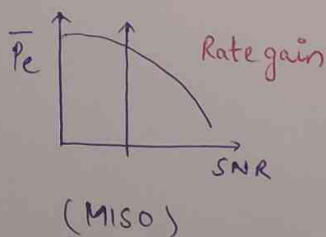
$$\text{Avg. BER, } \bar{P}_e = \int_0^{\infty} Q\left(\sqrt{\frac{2gP}{\sigma^2}}\right) f_g(g) dg$$

For high value of SNR,

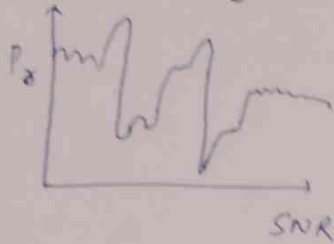
$$\bar{P}_e = \frac{1}{(\text{SNR})^L}$$



$L =$ diversity order.



Deep Fading Scenario :-



when Received signal strength almost reaches 0. is k/a deep fade scenario.

$$y = hx + n$$

$$SNR = \frac{|h|^2 P}{\sigma^2}$$

For this scenario,

$$SNR \ll 1$$

$$|h|^2 P < \sigma^2$$

$$|h|^2 < \frac{1}{(P/\sigma^2)}$$

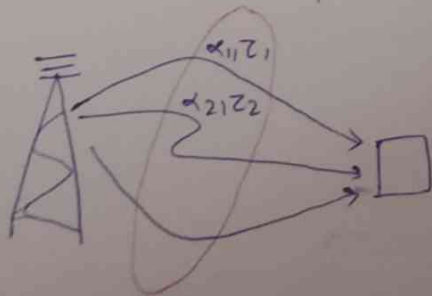
$$|h| \ll \frac{1}{\sqrt{P/\sigma^2}}$$

Prob. of deep fade -

$$P_r \left(|h| < \frac{1}{\sqrt{P/\sigma^2}} \right)$$

$$= \int_0^{\frac{1}{\sqrt{P/\sigma^2}}} f_{|h|}(h) dh$$

* Parametric Specifications of MPCs :-

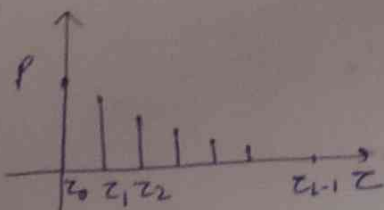


$$h(t) = \sum_{i=1}^L \alpha_i \delta(t - \tau_i)$$

$$P = E[|h(t)|^2]$$

$$= \sum |\alpha_i|^2 \delta(t)$$

PDP :- (Power Delay Profile)



↳ Max delay spread

$$\tau_{\max} = \tau_{L-1} - \tau_0$$

↳ Avg. delay spread

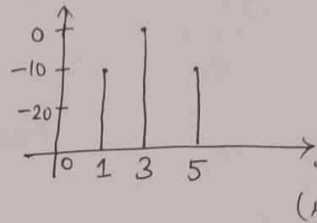
$$\bar{\tau} = \frac{\sum_{i=1}^L P_i \tau_i}{\sum_{i=1}^L P_i}$$

↳ RMS delay spread

$$\tau_{rms} = \sqrt{\frac{\sum P_i (\tau_i - \bar{\tau})^2}{\sum P_i}}$$

Ques

$P_0 = -20\text{dB}$	$\tau_0 = 0\mu\text{s}$
$P_1 = -10\text{dB}$	$\tau_1 = 1\mu\text{s}$
$P_2 = 0\text{dB}$	$\tau_2 = 3\mu\text{s}$
$P_3 = -10\text{dB}$	$\tau_3 = 5\mu\text{s}$



$P_0 = 0.01$
 $P_1 = 0.1$
 $P_2 = 1$
 $P_3 = 0.1$

$$\tau_{max} = 5 - 0 = 5\mu\text{s}$$

$$\bar{\tau} = \frac{0.01(0) + (0.1)(1) + (1)(3) + 0.1(5)}{1.21} = \frac{3.6}{1.21}$$

$$= 2.97\mu\text{s}$$

$$\tau_{rms} = 0.857\mu\text{s}$$

$\sqrt{2}$
 $\bar{\tau} - \tau_{rms}$ to $\bar{\tau} + \tau_{rms}$
 80% signal received

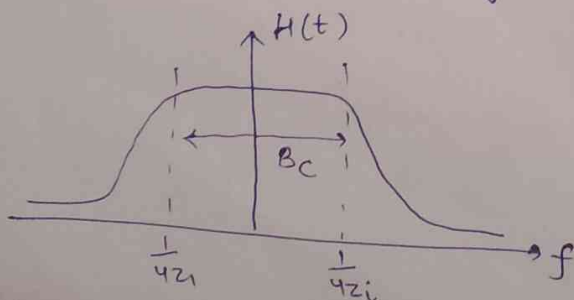
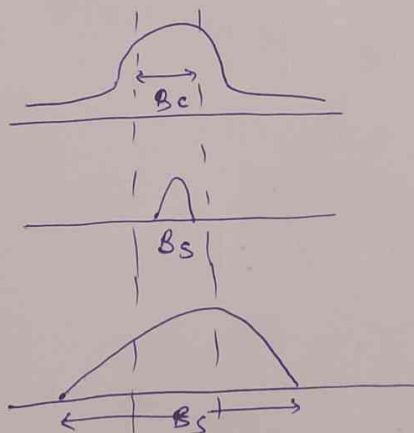
(*) Coherence Bandwidth

$y = hx + n$
 Multiply with h^{-1} ,
 $h^{-1}y = x + h^{-1}n$

$B_s < B_c \rightarrow$ No distortion
 $\rightarrow$ Flat Fading

$B_s > B_c \rightarrow$ Distortion / ISI
 $\rightarrow$ Freq Selective fading

$\rightarrow$ Channel BW over which behaviour of channel is constant or same.



$$\begin{aligned}
 H(f) &= \int h(t) e^{-j2\pi f t} dt \\
 &= \sum \alpha_i e^{-j2\pi f \tau_i}
 \end{aligned}$$

$$h(t) = \sum \alpha_i f(t - z_i)$$

$$H(f) = \int_0^{\infty} h(t) e^{-j2\pi f t} dt$$

$$= \sum \alpha_i \int f(t - z_i) e^{-j2\pi f t} dt$$

$$= \sum \alpha_i \int f(t - z) e^{-j2\pi f t} dt$$

$$H(f) = \alpha_i e^{-j2\pi f z_i}$$

$$f = 0$$

$$H(0) = \alpha_i$$

(Real)

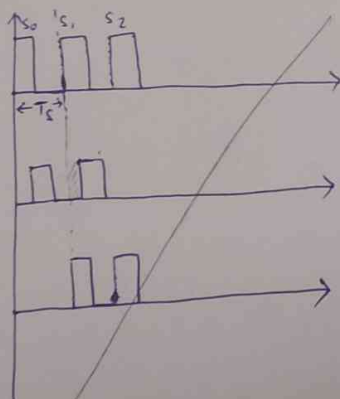
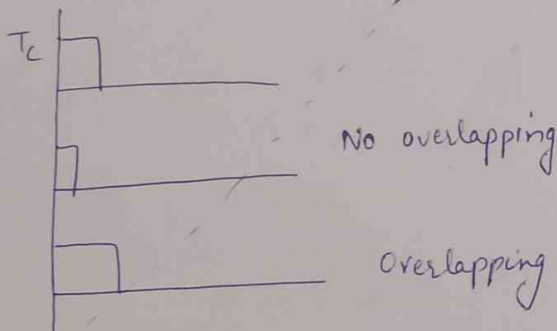
$$f = \frac{1}{4z_i}$$

$$H(f) = j\alpha_i$$

(Imag.)

$$B_c = \frac{1}{2z_i}$$

Coherence time :-
(σ_z)



$\sigma < T_s \rightarrow$ ISI
Overlapping

$\sigma \geq T_s \rightarrow$ NO ISI
No overlapping

$$B_s < B_c$$

$$\frac{1}{T_s} < \frac{1}{\sigma_z}$$

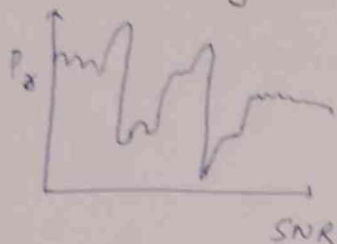
$$T_s > \sigma_z$$

$$B_s > B_c$$

$$T_s < \sigma_z$$

ISI
Freq. selective

Deep Fading Scenario :-



when Received signal strength almost reaches 0. is k/a deep fade scenario.

$$y = hx + n$$

$$SNR = \frac{|h|^2 P}{\sigma^2}$$

For this scenario,

$$SNR \lll 1$$

$$|h|^2 P < \sigma^2$$

$$|h|^2 < \frac{1}{\left(\frac{P}{\sigma^2}\right)^2}$$

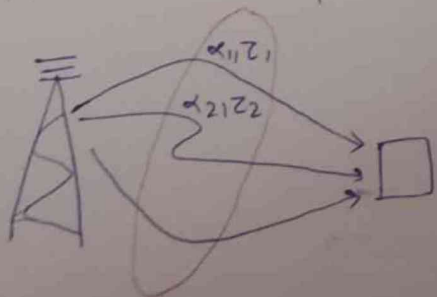
$$|h| \lll \frac{1}{\sqrt{\frac{P}{\sigma^2}}}$$

Prob. of deep fade -

$$P_r \left(|h| < \frac{1}{\sqrt{\frac{P}{\sigma^2}}} \right)$$

$$= \int_0^{\frac{1}{\sqrt{\frac{P}{\sigma^2}}}} f_{|h|}(h) dh$$

(*) Parametric Specifications of MPCs :-

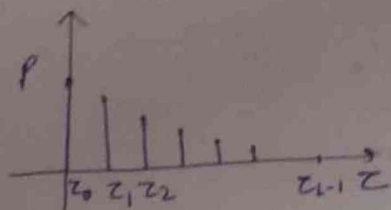


$$h(t) = \sum_{i=1}^L \alpha_i \delta(t - \tau_i)$$

$$P = E[|h(t)|^2]$$

$$= \sum |\alpha_i|^2 \delta(t)$$

PDP :- (Power Delay Profile)



↳ Max delay spread

$$\tau_{max} = \tau_{L-1} - \tau_0$$

↳ Avg. delay spread

$$\bar{\tau} = \frac{\sum_{i=1}^L P_i \tau_i}{\sum_{i=1}^L P_i}$$

$$h = \sum \alpha_i e^{-j2\pi f z_i} \cdot e^{j2\pi f \frac{v \cos \theta}{c} t}$$

$$h(t) = \underbrace{\sum \alpha_i e^{-j2\pi f z_i}}_{\text{flat fading}} \underbrace{e^{j2\pi f_d t}}_{\text{doppler}}$$

$$h(0) = h \quad (\text{flat fading})$$

$$h\left(\frac{1}{4f_d}\right) = jh$$

$$0 < t < \frac{1}{4f_d}$$

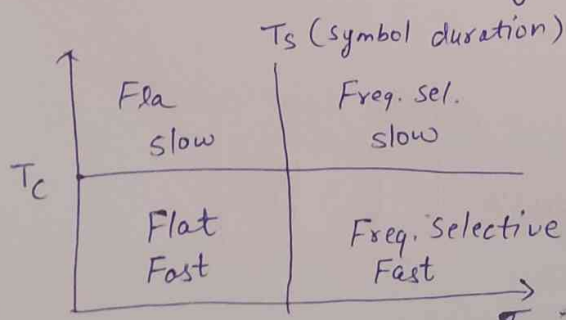
$$t = \frac{1}{4f_d} \rightarrow h \text{ will change}$$

This is coherence time

$$T_c = \frac{1}{2f_d} = \frac{1}{B_D}$$

$T_c >$ Interchanging estimation time
 $\rightarrow$ slow fading

$T_c <$ Interchanging estimation time
 $\rightarrow$ Fast Fading



Code Division Multiple Access (CDMA) :-

No concept of sim (sim change, operator change → not possible)
Every mobile assigned a unique code.

More security

But difficult to construct Billion of codes → drawback.

$$U_1 \rightarrow \alpha_1 \xrightarrow{C_1} [1, 1, 1, 1] \rightarrow [\alpha_1, \alpha_1, \alpha_1, \alpha_1]$$

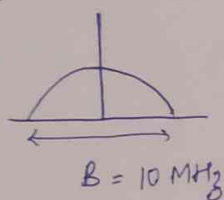
$$U_2 \rightarrow \alpha_2 \xrightarrow{C_2} [1, -1, -1, 1] \rightarrow [\alpha_2, -\alpha_2, -\alpha_2, \alpha_2]$$

$$y = [\alpha_1 + \alpha_2, \alpha_1 - \alpha_2, \alpha_1 - \alpha_2, \alpha_1 + \alpha_2] * C_1$$

$$= (\alpha_1 + \alpha_2) + (\alpha_1 - \alpha_2) + (\alpha_1 - \alpha_2) + (\alpha_1 + \alpha_2)$$

$$= 4\alpha_1$$

Note :-



$$T_s = \frac{1}{B} = 0.1 \mu s$$

⇓

$$\sigma = 2-3 \mu s \quad (\text{for outdoor envt})$$

$$T_s < \sigma$$

or

$$B_s > B_c \rightarrow \text{ISI}$$

(Limitation of single carrier modulation)

Solution → define multicarrier

i.e., divide total BW into n sub-carriers

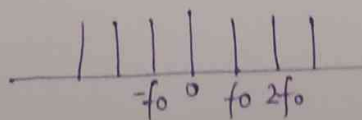
For each sub carrier, $B_{eff} = \frac{B}{N}$

MCM

N sub-carriers

N symbols

$(x_1, x_2, \dots, x_N)$



(each carrier will carry data so has a symbol)

$$s(t) = \sum_{k=1}^N X_k e^{j2\pi k f_0 t}$$

↓
Transmitted signal (total)

Assuming there is no noise or fading, Received Signal -

$$y(t) = \sum_{k=0}^{N-1} X_k e^{j2\pi k f_0 t}$$

For demodulation -

Demodulate l^{th} symbol $\xrightarrow{\text{fundamental Time Period}}$

$$\hat{y}(l) = \int_0^{\frac{1}{f_0}} y(t) e^{-j2\pi l f_0 t} dt$$

$$= \int_0^{\frac{1}{f_0}} \sum_{k=1}^N X_k e^{j2\pi f_0 (k-l)t} dt \quad | \quad k=l$$

=

Say $m(t)$ is message signal, (For SCM)

$$s(t) = m(t) \cos(2\pi f_c t)$$

$$y(t) = m(t) \cos(2\pi f_c t)$$

$$z(t) = y(t) \cos(2\pi f_c t)$$

$$= m(t) \left[\frac{1 + \cos(4\pi f_c t)}{2} \right]$$

Using LPF,

$$z(t) = \frac{m(t)}{2}$$

Drawback of MCM :-

For N no. of sub-carriers, it requires N modulators at transmitter end and N no. of local oscillators at receiver end which is not good.

Solution to MCM -

OFDM (Orthogonal Frequency Division Multiplexing)

OFDM

↳ IFFT / FFT $\rightarrow$ Rx end

↳ Tx end

↳ CP (Cyclic Prefix)

$$S(t) = \sum_{k=0}^{N-1} X_k e^{j2\pi k f_0 t}$$

$x(t) \rightarrow$ message signal $\rightarrow$ sampled at f_s

$$x(l) = x(lT_s) e^{j2\pi l f_0 T_s} \quad \{ l^{\text{th}} \text{ sample value} \}$$

$$T_s = \frac{1}{f_s}$$

$f_s \rightarrow$ max. msg signal freq.

$$f_s = \frac{B}{N}$$

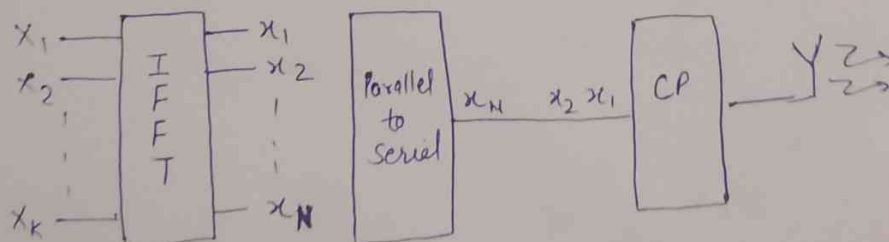
$$T_s = \frac{N}{B}$$

$$x(l) = x(lT_s) e^{j2\pi l \frac{N}{B} f_0}$$

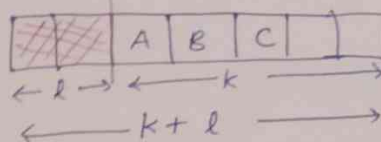
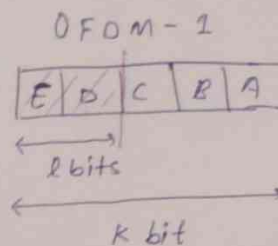
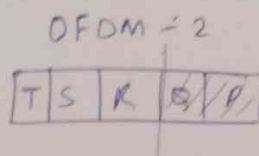
$$x(l) = X_l e^{j2\pi l \frac{N}{B} f_0}$$

$$x(t) = \sum_{l=1}^N X_l e^{j2\pi l \frac{N}{B} f_0}$$

IFFT

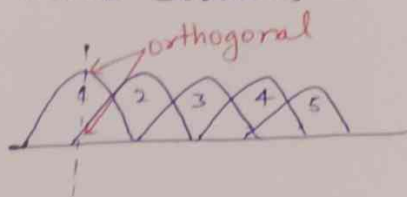


T_x Block diagram



This is k/a
Cyclic Prefix

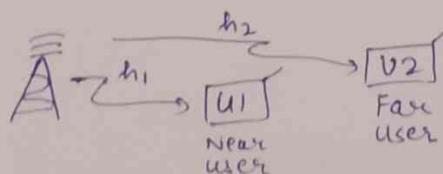
N sub-carriers $\rightarrow$ all are orthogonal to each other
FD $\rightarrow$ divide user based on frequency.



Drawback - BW utilization is not good
i.e., Spectral efficiency is not optimum

$\downarrow$
Solution $\rightarrow$ NOMA (Not Orthogonal Multiple Access)

NOMA



$$\begin{aligned} x_1 &\rightarrow U_1 & p_1 &\rightarrow U_1 \\ x_2 &\rightarrow U_2 & p_2 &\rightarrow U_2 \end{aligned}$$

$$x = \sqrt{p_1} x_1 + \sqrt{p_2} x_2$$

Assume, $h_1 > h_2$ (h_1 is good link, has less losses)
 $\Rightarrow p_1 < p_2$ (More power to signal 2)

$$\begin{aligned} y_1 &= h_1 (\sqrt{p_1} x_1 + \sqrt{p_2} x_2) + n_1 \\ y_2 &= h_2 (\sqrt{p_1} x_1 + \sqrt{p_2} x_2) + n_2 \end{aligned}$$

$n_1, n_2 \rightarrow$ AWGN Noise

$$n_1, n_2 \approx \mathcal{N}(0, \sigma^2)$$

$y_1, y_2 \rightarrow$ received signal.

⇒ Successive Interference Cancellation (SIC)

Decode $\hat{x}_2$

← At User 1

$$\hat{y}_1 = y_1 - h_1 \sqrt{P_2} \hat{x}_2$$

If decoded perfectly,

$$\hat{y}_1 = h_1 \sqrt{P_1} x_1 + n_1$$

Now,

$$SNR = \frac{|h_1|^2 P_1}{\sigma^2}$$

$$\Rightarrow SINR = \frac{|h_2|^2 P_2}{\underbrace{|h_2|^2 P_1}_{\text{Interference power}} + \underbrace{\sigma^2}_{\text{noise power}}}$$

(Not the average SNR)

For BPSK Mod.:-

$$P_e = Q(\sqrt{SNR})$$

$$P_{e,1}^T = P_{e1 \rightarrow 2} + (1 - P_{e1 \rightarrow 2}) P_{e,1} \quad (\text{for user 1})$$

For user 2, we can get P_e directly.

$$\text{Now, } P_{e1 \rightarrow 2} = Q(\sqrt{SINR_{1-2}})$$

$$SINR_{1-2} = \frac{|h_1|^2 P_2}{|h_1|^2 P_1 + N_0}$$